



Guangdong Technion

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GTIIT

CALCULUS 1 - M

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LECTURE 17 - DECEMBER 9TH

TECHNIQUES OF INTEGRATION

INTEGRATION BY PARTS

Every differentiation rule has a **corresponding integration rule**:

- Substitution Rule for integration $\leftrightarrow$ Chain Rule for differentiation.
- Product Rule for differentiation $\leftrightarrow$ Integration by parts.

The Product Rule states that if f and g are differentiable functions, then

$$(f g)' = f' g + f g'$$

Integrate both sides: on the left hand side, the integral and the derivative are inverse processes:

$$f(x) g(x) = \int (f(x) g(x))' dx = \int [f'(x)g(x) + f(x)g'(x)] dx$$

INTEGRATION BY PARTS

$$f(x) g(x) = \int \left(f(x) g(x) \right)' dx = \int \left[f'(x)g(x) + f(x)g'(x) \right] dx$$

So, we can solve one of both terms of the right hand side:

$$\int f(x)g'(x)dx = f(x) g(x) - \int f'(x)g(x)dx (*)$$

Write $u = f(x)$ and $v = g(x)$, then the differentials are

$$du = f'(x)dx \text{ and } dv = g'(x)dx$$

Replace into $(*)$ to get

$$\int u dv = u v - \int v du$$

INTEGRATION BY PARTS

Ex. 1. Solve $\int x \sin x \, dx$.

By Integration **by Parts**, choose:

- $u = x \Rightarrow 1 = du$
- $dv = \sin x \Rightarrow v = -\cos x$

Then

$$\int x \sin x \, dx$$

$$= \int u \, dv = u \, v - \int v \, du$$

$$= -x \cos x + \int \cos x \cdot 1 \, dx = -x \cos x + \sin x + C \quad \checkmark$$

HOW TO CHOOSE u AND v IN INTEGRATION BY PARTS

- Our aim in using integration by parts is to obtain a **simpler integral** than the one we started with.
- In Example 1, if instead of we did, we had chosen $u = \sin x$ and $dv = x$, then

$$du = \cos x \text{ and } v = \frac{1}{2}x^2$$

so integration by parts gives **a more difficult formula to solve**, which **makes things worse**:

$$\int x \sin x \, dx = \frac{1}{2}x^2 \sin x - \frac{1}{2} \int x^2 \cos x \, dx \leftarrow \text{not useful}$$

- Idea: choose u as a function that becomes simpler when differentiated, as long as dv can be readily integrated to give v .

INTEGRATION BY PARTS

Example 2. Solve $\int \arctan(x) dx$

Solution. Choose • $u = \arctan(x) \Rightarrow du = \frac{1}{1+x^2} dx$

• $1 dx = dv \Rightarrow v = x$:

$$\int \underbrace{\arctan(x)}_u \cdot \underbrace{1 \cdot dx}_{dv}$$

$$= \int u dv = uv - \int v du$$

$$= x \arctan(x) - \int \frac{x}{1+x^2} dx = x \arctan(x) - \frac{1}{2} \ln(1+x^2) + C \checkmark$$

INTEGRATION BY PARTS

Example 3. a) Solve $\int \sin(x)e^x dx$

Solution. In this case, there is no hierarchy in choose the exponential or the trigonometric factor as u :

- $u = e^x \Rightarrow du = e^x dx$
- $\sin(x) dx = dv \Rightarrow v = -\cos(x)$:

$$\int \underbrace{e^x}_u \underbrace{\sin(x)dx}_{dv} = \boxed{\int u dv = uv - \int v du} = -e^x \cos(x) + \int e^x \cos(x) dx$$

This is not better than the initial integral; try again:

$$\begin{aligned} &= -e^x \cos(x) + \int \underbrace{\cos(x)}_u \cdot \underbrace{e^x dx}_{dv} \\ &= -e^x \cos(x) + e^x \sin(x) - \int e^x \sin(x) dx \end{aligned}$$

INTEGRATION BY PARTS:

This seems to be the same as the beginning, but, it allows us to solve:

$$\int \sin(x)e^x dx = -e^x \cos(x) + e^x \sin(x) - \int \sin(x)e^x dx$$

$$\Rightarrow 2 \left[\int \sin(x)e^x dx \right] = -e^x \cos(x) + e^x \sin(x)$$

$$\Rightarrow \int \sin(x)e^x dx = \frac{1}{2}e^x [-\cos(x) + \sin(x)] + C \checkmark$$

EVALUATE DEFINITE INTEGRALS BY PARTS

By the Fundamental Theorem of Calculus:

$$\int_a^b u \, dv = u v \Big|_a^b - \int_a^b v \, du$$

$$\int_a^b f(x)g'(x)dx = f(x)g(x) \Big|_a^b - \int_a^b g(x)f'(x)dx$$

INTEGRATION BY PARTS

3. b) Solve $\int_0^{\pi} \sin(x)e^x dx$

Solution. Use the antiderivative from 3. a)

$$\begin{aligned}\int_0^{\pi} \sin(x)e^x dx &= \left[\frac{1}{2}e^x [-\cos(x) + \sin(x)] + C \right] \Big|_0^{\pi} \\&= \frac{1}{2}e^{\pi} [-\cos(\pi) + \sin(\pi)] + C - \left\{ \frac{1}{2}e^0 [-\cos(0) + \sin(0)] + C \right\} \\&= \frac{1}{2}(e^{\pi} + 1) \checkmark\end{aligned}$$

INTEGRATION BY PARTS: LIATE RULE

The **LIATE** Rule consist of choosing as u the function which appears first in the following list:

- L Logarithmic functions $\ln(x)$, $\log_b(x)$
- I Inverse trigonometric and hyperbolic functions
 $\arctan(x)$, $\operatorname{arsinh}(x)$, ...
- A Algebraic functions: compositions of rational powers
- T Trigonometric and hyperbolic functions
- E Exponential functions e^x , a^x

Ex. 4. Prove for $n \geq 0$

$$\int \sin^n(x) \cos(x) dx = \frac{1}{n+1} \sin^{n+1}(x) + C$$

Solution. By substitution, $u = \sin(x)$, $du = \cos(x) dx$:

$$\begin{aligned} \int \sin^n(x) \cos(x) dx &= \int u^n du \\ &= \frac{1}{n+1} u^{n+1} + C = \frac{1}{n+1} \sin^{n+1}(x) + C \checkmark \end{aligned}$$

REDUCTION FORMULA

Ex 5. Prove $\forall n \geq 2, n \in \mathbb{Z}$

$$\int \sin^n(x) dx = -\frac{1}{n} \sin^{n-1}(x) \cos(x) + \frac{n-1}{n} \int \sin^{n-2}(x) dx$$

The reduction formula is useful because, in each step, it reduces in 2 units the exponent, so by using it repeatedly, gives an expression of $\int \sin^n(x) dx$ in terms of

$\int \sin(x) dx$, if n is odd, or $\int \sin^0(x) dx = \int 1 dx$, if n is even.

REDUCTION FORMULA

Proof. Let

- $u = \sin^{n-1}(x)$
- $du = (n-1) \sin^{n-2}(x) \cos(x) dx$
- $dv = \sin(x)$
- $v = -\cos(x)$.

By Integration by Parts: $\int u dv = uv - \int v du$:

$$\int \underbrace{\sin^{n-1}(x)}_u \underbrace{\sin(x) dx}_{dv} = -\sin^{n-1}(x)\cos(x) + (n-1) \int \sin^{n-2}(x)\cos^2(x) dx$$

Since $\cos^2(x) = 1 - \sin^2(x)$, replace this:

$$= -\sin^{n-1}(x)\cos(x) + (n-1) \int \sin^{n-2}(x) (1 - \sin^2(x)) dx$$

$$= -\sin^{n-1}(x)\cos(x) + (n-1) \int \sin^{n-2}(x) dx - (n-1) \int \sin^n(x) dx$$

The last is $(n - 1)$ times the left hand side, so, we place it to the right, and get n times the initial integral.

$$n \int \sin^n(x) dx = -\sin^{n-1}(x)\cos(x) + (n - 1) \int \sin^{n-2}(x) dx$$

To finish, divide by n :

$$\int \sin^n(x) dx = -\frac{1}{n} \sin^{n-1}(x)\cos(x) + \frac{(n - 1)}{n} \int \sin^{n-2}(x) dx \checkmark$$

TRIGONOMETRIC INTEGRALS

Recall

$$\sin^2(x) = \frac{1 - \cos(2x)}{2},$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

Ex. 6. Solve

$$\int \sin^4(x) dx$$

Solution. We can use the reduction formula or the previous one:

$$\begin{aligned}\int \sin^4(x) dx &= \int [\sin^2(x)]^2 dx = \int \left(\frac{1 - \cos(2x)}{2} \right)^2 dx \\&= \frac{1}{4} \int (1 - 2\cos(2x) + \cos^2(2x)) dx \\&= \frac{1}{4} \int \left(1 - 2\cos(2x) + \frac{1}{2}(1 + \cos(4x)) \right) dx = \frac{1}{4} \int \left(\frac{3}{2} - 2\cos(2x) + \frac{1}{2}\cos(4x) \right) dx \\&= \frac{1}{4} \left(\frac{3}{2}x - \sin(2x) + \frac{1}{8}\sin(4x) \right) + C\end{aligned}$$

Ex. 7. Solve $\int \sin^5(x) \cos^2(x) dx$.

Solution. Before work with the integral, try to rewrite the expression using **Pythagorean Theorem**:

$$\sin^5(x) \cos^2(x) = (\sin^2(x))^2 \cos^2(x) \sin(x) = (1 - \cos^2(x))^2 \cos^2(x) \sin(x)$$

Next, consider the integral and substitute

$$\Rightarrow u = \cos(x) \Rightarrow du = -\sin(x) dx.$$

$$\begin{aligned} \int \sin^5(x) \cos^2(x) dx &= \int (1 - \cos^2(x))^2 \cos^2(x) \sin(x) dx \\ &= \int (1 - u^2)^2 u^2 (-du) = - \int (u^2 - 2u^4 + u^6) du = -\frac{1}{3}u^3 + \frac{2}{5}u^5 - \frac{1}{7}u^7 + C \\ &= -\frac{1}{3} \cos^3 x + \frac{2}{5} \cos^5 x - \frac{1}{7} \cos^7 x + C \quad \checkmark \end{aligned}$$

TRIGONOMETRIC INTEGRALS

Ex. 8. a) Solve $\int \tan(x) dx$

$$\begin{aligned}\text{Solution. } \int \tan(x) dx &= \int \frac{\sin(x)}{\cos(x)} dx = \int \frac{-du}{u} \\ &= -\ln(\cos(x)) + C \\ &= \ln\left(\frac{1}{\cos(x)}\right) + C = \boxed{\ln |\sec(x)| + C} \checkmark\end{aligned}$$

If by hypothesis $x \in [a, b]$ where $\cos(x) > 0$ then

$$\boxed{\int \tan(x) dx = \ln(\sec(x)) + C} \checkmark$$

Ex. 8. b) Prove

$$\tan^2(x) = \sec^2(x) - 1 \quad \text{and} \quad \frac{d}{dx} [\sec(x)] = \sec(x) \tan(x)$$

c) Solve $\int \tan^5(x) \sec^7(x) dx$.

Solution b)

$$\begin{aligned} \bullet \tan^2(x) &= \frac{\sin^2(x)}{\cos^2(x)} = \frac{1 - \cos^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} - 1 \\ &= \sec^2(x) - 1 \end{aligned}$$

$$\begin{aligned} \bullet \frac{d}{dx} [\sec(x)] &= \frac{d}{dx} \left[\frac{1}{\cos(x)} \right] = -\frac{1}{\cos^2(x)} (-\sin(x)) \\ &= \frac{1}{\cos(x)} \frac{\sin(x)}{\cos(x)} = \sec(x) \tan(x) \end{aligned}$$

$$\text{SOLUTION C) } \int \tan^5(x) \sec^7(x) dx.$$

Use the identity and substitute

$$u = \sec(x) \implies du = \sec(x) \tan(x) dx$$

For the given integral, we have then

$$\begin{aligned} \int \tan^5(x) \sec^7(x) dx &= \int \tan^4(x) \sec^6(x) \sec(x) \tan(x) dx \\ &= \int \left[\underbrace{\sec^2(x) - 1}_{=u^2-1} \right]^2 \underbrace{\sec^6(x)}_{u^6} \underbrace{\sec(x) \tan(x) dx}_{=du} \\ &= \int \left[(u^4 - 2u^2 + 1) u^6 \right] du = \int \left[(u^{10} - 2u^8 + u^6) \right] du = \frac{1}{11} u^{11} - \frac{2}{9} u^9 + \frac{1}{7} u^7 + C \\ &= \frac{1}{11} \sec^{11}(x) - \frac{2}{9} \sec^9(x) + \frac{1}{7} \sec^7(x) + C \checkmark \end{aligned}$$

EX. 9. PROVE

$$\int \sec(x) dx = \ln \left| \sec(x) + \tan(x) \right| + C \quad \checkmark$$

Proof. Derivate the right side $F(x) = \ln \left| \sec(x) + \tan(x) \right|$:

$$\frac{d}{dx} F(x) = \frac{1}{\left(\sec(x) + \tan(x) \right)} \left(\frac{\sin(x)}{\cos^2(x)} + \frac{1}{\cos^2(x)} \right)$$

$$= \frac{1}{\left(\frac{1}{\cos(x)} + \frac{\sin(x)}{\cos(x)} \right)} \left(\frac{\sin(x)}{\cos^2(x)} + \frac{1}{\cos^2(x)} \right) = \frac{1}{\cos(x)} = \sec(x) \quad \checkmark$$

$$\int \tan^m(x) \sec^{2k}(x) dx : k \geq 2, m \geq 1$$

- 1 For k at least 2, save a factor of $\sec^2(x)$.
- 2 Use $\sec^2(x) = 1 + \tan^2(x)$ to express the remaining factors in terms of $\tan^2(x)$:

$$\begin{aligned} \int \tan^m(x) \sec^{2k}(x) dx &= \int \tan^m(x) (\sec^2(x))^{k-1} \sec^2(x) dx \\ &= \int \tan^m(x) (1 + \tan^2(x))^{k-1} \sec^2(x) dx \end{aligned}$$

- 3 Substitute $u = \tan(x) \Rightarrow du = \sec^2(x) dx$:

$$\int \tan^m(x) \sec^{2k}(x) dx = \int u^m (1 + u^2)^{k-1} du \quad \checkmark$$

ESTRATEGY FOR EVALUATING

$$\int \tan^{2k+1}(x) \sec^n(x) dx : k \geq 1, n \geq 1$$

- 1 For k at least 1, save a factor of $\sec(x) \tan(x)$.
- 2 Use $\tan^2(x) = \sec^2(x) - 1$ to express the remaining factors in terms of $\sec(x)$:

$$\begin{aligned} \int \tan^{2k+1}(x) \sec^n(x) dx &= \int (\tan^2(x))^k \sec^{n-1}(x) \sec(x) \tan(x) dx \\ &= \int (\sec^2(x) - 1)^k \sec^{n-1}(x) \sec(x) \tan(x) dx \end{aligned}$$

- 3 Substitute $u = \sec(x) \Rightarrow du = \sec(x) \tan(x) dx$:

$$\int \tan^{2k+1}(x) \sec^n(x) dx = \int (u^2 - 1)^k u^{n-1} du \quad \checkmark$$

ESTRATEGY FOR EVALUATING

$$\int \tan^m(x) \sec^n(x) dx : m, n \geq 1$$

Other cases: use similar strategies, integration by parts, substitution

Need the property:

$$\int \sec(x) dx = \ln \left| \sec(x) + \tan(x) \right| + C \checkmark$$

ESTRATEGY FOR EVALUATING

$$\int \sin(mx) \cos(nx) dx : m, n \geq 1 .$$

Use the identities:

$$\sin(A) \cos(B) = \frac{1}{2} \left[\sin(A - B) + \sin(A + B) \right] .$$

$$\sin(A) \sin(B) = \frac{1}{2} \left[\cos(A - B) - \cos(A + B) \right] .$$

$$\cos(A) \cos(B) = \frac{1}{2} \left[\cos(A - B) + \cos(A + B) \right] .$$

TRIGONOMETRIC INTEGRALS

Ex. 10. Solve $\int \sin(4x) \cos(5x) dx$

Solution.
$$\begin{aligned}\int \sin(4x) \cos(5x) dx &= \int \frac{1}{2} [\sin(-x) + \sin(9x)] dx \\ &= \int \frac{1}{2} [-\sin(x) + \sin(9x)] dx \\ &= \frac{1}{2} \left[\cos(x) - \frac{1}{9} \cos(9x) \right] + C \checkmark\end{aligned}$$

TRIGONOMETRIC SUBSTITUTIONS

<i>Expression</i>	<i>Substitution</i>	<i>Identity</i>
$\sqrt{a^2 - x^2}$	$x = a \sin(\theta), -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2(\theta) = \cos^2(\theta)$
$\sqrt{a^2 + x^2}$	$x = a \tan(\theta), -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2(\theta) = \sec^2(\theta)$
$\sqrt{x^2 - a^2}$	$x = a \sec(\theta),$ $0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\sec^2(\theta) - 1 = \tan^2(\theta)$

TRIGONOMETRIC SUBSTITUTIONS

Ex. 11. a) Solve

$$\int \cot^2(t) dt = \int (\csc^2(t) - 1) dt = -\cot(t) - t + C.$$

b) Solve $\int \frac{\sqrt{9-x^2}}{x^2} dx$

Solution a) Exercise.

b) Let $x = 3 \sin(t) \Rightarrow dx = 3 \cos(t) dt$ for $0 < t \leq \frac{\pi}{2}$:

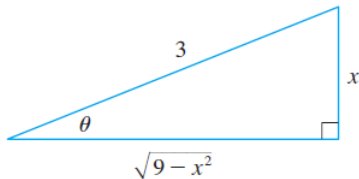
$$\sqrt{9-x^2} = \sqrt{9-9\sin^2(t)} = \sqrt{9\cos^2(t)} = 3|\cos t| = 3\cos(t)$$

$$\Rightarrow \int \frac{\sqrt{9-x^2}}{x^2} dx = \int \frac{3\cos(t)}{9\sin^2(t)} 3\cos(t) dt = \int \frac{\cos^2(t)}{\sin^2(t)} dt$$

$$= -\cot(t) - t + C \checkmark$$

SUBSTITUTE BACK TO THE ORIGINAL VARIABLE

- The opposite side and the hypotenuse have lengths x and 3 , respectively, then the adjacent side has length of $3 \cos(t)$:



$$x = 3 \sin(t) \Rightarrow \ell = \text{hypotenuse} = 3 \Rightarrow 3 \cos(t) = \sqrt{9 - x^2}$$

- Then, **cotangent of t** can be solved:

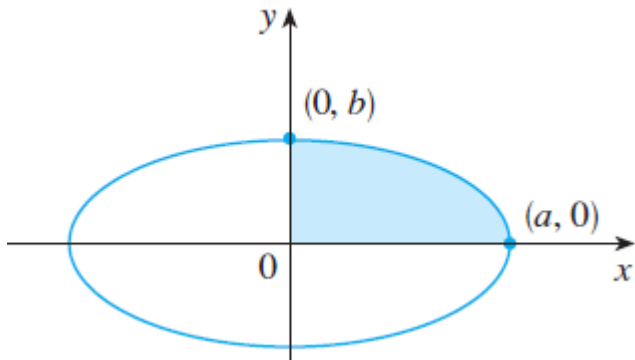
$$\cot(t) = \frac{3 \cos(t)}{3 \sin(t)} = \frac{\sqrt{9 - x^2}}{x}$$

Hence, the given integral is:

$$\Rightarrow \int \frac{\sqrt{9 - x^2}}{x^2} dx = -\frac{\sqrt{9 - x^2}}{x} - \arcsin\left(\frac{x}{3}\right) + C \quad \checkmark$$

12. FIND THE AREA ENCLOSED BY THE ELLIPSE

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 : a, b > 0$$



FIND THE AREA ENCLOSED BY THE ELLIPSE

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 : a, b > 0$$

Solution. • Solve the equation of the ellipse for $y \geq 0$:

$$y = \frac{b}{a} \sqrt{a^2 - x^2}$$

- Because the ellipse is symmetric with respect to both axes, the total area is **4 times the area in the first quadrant**:

$$y = f(x) = \frac{b}{a} \sqrt{a^2 - x^2} : 0 \leq x \leq a$$

- Hence, the area is

$$A = \int_0^a 4 \frac{b}{a} \sqrt{a^2 - x^2} dx$$

FIND THE AREA ENCLOSED BY THE ELLIPSE

- Substitute $x = a \sin(\theta) \Rightarrow dx = a \cos(\theta) d\theta$ for $0 \leq \theta \leq \frac{\pi}{2}$

$$\bullet \sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2(\theta)} = \sqrt{a^2 \cos^2(\theta)}$$

$$= a |\cos(\theta)| = a \cos(\theta)$$

$$\Rightarrow A = \int_0^a 4 \frac{b}{a} \sqrt{a^2 - x^2} dx = 4 \frac{b}{a} \int_0^{\pi/2} a \cos(\theta) \cdot a \cos(\theta) d\theta$$

$$= 4ab \int_0^{\pi/2} \cos^2(\theta) d\theta = 4ab \int_0^{\pi/2} \frac{1}{2} [1 + \cos(2\theta)] d\theta$$

$$= 2ab \left[\theta + \frac{1}{2} \sin(2\theta) \right]_0^{\pi/2} = 2ab \left(\frac{\pi}{2} + 0 - 0 \right) = \pi ab \checkmark$$

TRIGONOMETRIC SUBSTITUTIONS

Ex. 13. Solve $\int \frac{1}{\sqrt{x^2 - a^2}} dx : a > 0$

Solution Let $x = a \sec(\theta) \Rightarrow dx = a \sec(\theta) \tan(\theta) d\theta$ for $0 < \theta \leq \frac{\pi}{2}$:

$$\sqrt{x^2 - a^2} = \sqrt{a^2(\sec^2(\theta) - 1)} = \sqrt{a^2 \tan^2(\theta)} = a |\tan \theta| = a \tan \theta$$

$$\Rightarrow \int \frac{1}{\sqrt{x^2 - a^2}} dx = \int \frac{a \sec(\theta) \tan(\theta) d\theta}{a \tan(\theta)} = \int \sec(\theta) d\theta$$

$$= \ln \left| \sec(\theta) + \tan(\theta) \right| + C = \ln \left(\frac{x + \sqrt{x^2 - a^2}}{a} \right) + C$$

$$= \ln \left(x + \sqrt{x^2 - a^2} \right) + \tilde{C} \checkmark$$

INTEGRATION OF RATIONAL FUNCTION BY PARTIAL FRACTIONS

We will study how to integrate any rational function, a ratio of polynomials, by expressing it as a sum of simpler fractions, called partial fractions, that we already know how to integrate

INTEGRATION OF RATIONAL FUNCTION BY PARTIAL FRACTIONS

Example: Solve

$$\int \frac{x+5}{x^2+x-2} dx$$

Solution. Write the integrand, factorizing the denominator according to **the real roots**:

$$x^2 + x - 2 = (x-1)(x+2)$$

Then,

$$\frac{x+5}{x^2+x-2} = \frac{x+5}{(x-1)(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x+2)}$$

This is true for **$A = 2$** , **$B = -1$** , so

$$\frac{x+5}{x^2+x-2} = \frac{2}{(x-1)} - \frac{1}{(x+2)}$$

INTEGRATION OF RATIONAL FUNCTION BY PARTIAL FRACTIONS

Hence, the integral is easy:

$$\int \frac{x+5}{x^2+x-2} dx$$

$$= \int \frac{2}{(x-1)} dx - \int \frac{1}{(x+2)} dx = 2 \ln(x-1) - \ln(x+2) \checkmark$$